

Empirical Proof Record: VALIDATED

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Proof ID: 9e69e1db820fba5d982a240866e547e1...

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1. Claim

Under sustained load, the substrate stays cognitively alive on real input: (for every cycle that produced a concept set, $\Phi \geq 0.05$). Empty-concept cycles are excluded — with nothing to integrate, $\Phi=0$ is correct, not a defect; their rate is reported separately. The floor is the lowest- Φ real-input cycle across the run.

- **Operationalisation:** Stream 3 passes over 8 substantive stimuli through Kimera’s entity target (Takwin); read the per-cycle “phi” and “concepts”. Partition cycles into real-input (concepts non-empty) and empty-input. $\text{phi_floor} = \min \Phi$ over real-input cycles; the LTL invariant holds iff that $\text{floor} \geq \varphi_floor$. empty_input_rate and the over-all-cycles phi_floor_strict are reported alongside.
- **Threshold:** $\text{phi_floor} \geq 0.05$ phi
- **H0:** $H0: \min \Phi < 0.05$ on real-input cycles — Φ collapses below the floor despite the input producing concepts
- **H1:** $H1: \min \Phi \geq 0.05$ on real-input cycles — the substrate stays cognitively alive whenever it has something to integrate

2. Verdict

VALIDATED — observed 0.6156636348270582

Φ floor 0.6157 over 21 real-input cycles (3 passes $\times$ 8 stimuli); mean Φ 0.6372, band [0.6157, 0.6561], stdev 0.0123; non-collapse rate 100.0%; 3 empty-input cycles excluded (12.5%; strict floor over all cycles 0.0000); 0 cycles produced no Φ ; worst real cycle: stimulus 4 @ cycle 20 = 0.6157

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.phi_stability	phi_floor	0.6156636348270582	—	ophamin 0.82.0

4. Pre-registration

- Registered at: 2026-05-21T23:41:37.833041+00:00
- Config hash: fc0d6fa2b433540d0fc987795a9290da...
- Data hash: f36c007933b4b5223d517967d58c7b80...

5. Signature

c7ea620639b46d64637b9e2d9c1aa378957f94acd7f149adcacf08fea3fc71ec